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# Speech Enhancement Based on the General Transfer Function GSC and Postfiltering

**Sharon Gannot and Israel Cohen**

## ABSTRACT

In speech enhancement applications microphone array postfilteing allows additional reduction of noise components at a beamformer output. Among microphone array structures the recently proposed General Transfer function Generalized Sidelobe Canceller (TF-GSC) has shown impressive noise reduction abilities in a directional noise field, while still maintaining low speech distortion. However, in a diffused noise field less significant noise reduction is obtainable. The performance is even further degraded when the noise signal is nonstationary. In this contribution we propose three postfiltering methods for improving the performance of microphone arrays. Two of which are based on single–channel speech enhancers and making use of recently proposed algorithms concatenated to the beamformer output. The third is a multi--channel speech enhancer which exploits noise--only components constructed within the TF-GSC structure. This work concentrates on the assessment of the proposed postfiltering structures. An extensive experimental study, which consists of both objective and subjective evaluation in various noise fields, demonstrates the advantage of the multi—channel postfiltering compared to the single–channel techniques.